

NORFOLK NAS, VA, USA

WMO: 723085

Lat: 36.937N

Lon: 76.289W

Elev: 17

StdP: 14.69

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13750

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.5	26.8	2.5	6.3	26.5	7.3	8.0	30.5	28.5	36.6	25.9	36.8	12.6	330	0.474

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.1	93.9	77.0	91.5	76.1	89.7	75.5	79.7	89.5	78.4	87.7	77.5	86.2	8.7	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.1	141.1	84.0	75.6	134.0	83.0	74.9	130.8	82.3	43.1	89.6	41.8	88.6	40.8	86.1	86.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.1	21.5	19.0	17.8	98.7	5.2	2.3	14.1	100.3	11.1	101.7	8.2	103.0	4.5	104.7
			14.5	82.3	4.7	1.8	11.2	83.6	8.4	84.7	5.8	85.7	2.4	87.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	62.1	42.4	44.6	50.8	60.8	69.1	77.1	80.9	79.4	74.4	64.4	53.7	46.9
	DBStd	15.58	10.37	9.23	9.37	8.61	7.39	5.75	4.38	4.17	5.03	7.44	8.19	9.08
	HDD50	823	284	201	106	10	0	0	0	0	1	52	169	
	HDD65	2974	703	573	449	183	45	2	0	0	104	349	564	
	CDD50	5247	47	50	132	333	591	812	957	912	732	446	162	73
	CDD65	1926	1	2	10	57	171	365	492	447	284	84	9	4
	CDH74	16709	2	10	72	456	1299	3303	5107	4159	1814	447	31	9
	CDH80	5925	0	0	8	120	412	1257	2095	1474	471	87	1	0
Wind	WSAvg	9.1	10.0	10.0	10.6	10.3	9.0	8.1	7.6	7.5	8.9	8.9	9.2	9.2
Precipitation	PrecAvg	41.1	3.4	3.3	3.5	3.0	3.0	3.5	4.5	4.4	4.0	2.9	2.6	3.0
	PrecMax	67.8	8.9	8.1	9.1	6.7	9.3	9.6	12.2	11.8	18.2	9.5	6.3	6.6
	PrecMin	21.8	0.9	0.9	0.8	0.5	0.6	0.8	1.3	0.9	0.5	0.2	0.1	0.5
	PrecStd	10.6	1.5	1.5	1.7	1.7	1.7	1.9	2.3	2.6	4.1	2.0	1.6	1.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	72.7	75.2	81.1	88.1	91.4	96.6	97.4	95.9	91.5	86.0	77.9	74.9
		MCWB	63.1	63.4	64.3	68.3	73.6	76.2	78.0	77.5	74.5	71.9	65.9	65.6
	2%	DB	68.0	69.6	75.3	83.4	88.1	92.6	94.3	92.4	88.2	82.3	73.3	70.2
		MCWB	60.6	59.1	61.6	66.2	72.1	75.4	78.0	77.1	74.2	70.4	65.0	63.3
	5%	DB	63.7	64.9	71.2	79.8	84.6	90.0	91.9	90.1	85.1	79.1	70.1	65.8
		MCWB	57.7	56.7	59.7	64.8	70.9	74.9	77.1	76.2	73.2	69.4	62.7	60.8
	10%	DB	59.0	60.5	66.5	75.0	81.4	87.2	89.8	87.6	82.2	75.5	66.4	61.7
		MCWB	53.1	52.6	57.7	63.3	69.3	73.8	76.6	75.6	72.5	67.6	60.2	56.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	65.7	65.3	67.0	71.5	77.0	79.9	82.2	80.6	78.1	75.9	69.8	68.1
		MCDB	70.9	72.3	76.2	81.9	86.2	89.8	89.6	90.2	85.5	81.6	74.3	72.9
	2%	WB	62.1	62.1	64.5	69.3	74.5	78.0	80.2	79.1	76.8	73.3	66.9	64.2
		MCDB	66.6	68.0	71.7	79.4	84.0	88.1	89.8	88.4	83.3	78.9	71.5	68.7
	5%	WB	58.3	58.2	62.1	67.2	72.7	76.7	79.0	78.0	75.7	71.1	64.1	61.3
		MCDB	63.5	64.2	69.2	76.6	81.7	86.2	88.8	86.6	81.6	77.0	68.5	65.5
	10%	WB	53.6	53.0	58.7	64.7	71.0	75.5	77.9	77.1	74.6	68.8	61.2	57.1
		MCDB	58.3	59.3	65.9	73.2	79.1	84.0	87.1	84.9	80.3	74.8	66.0	61.5
Mean Daily Temperature Range	5% DB	MDBR	14.4	15.5	16.2	17.4	15.6	14.6	14.1	13.1	12.4	14.2	14.8	14.0
		MCDBR	21.2	23.4	23.4	22.9	19.8	18.3	17.3	16.3	16.2	17.2	19.1	19.0
		MCWBR	16.3	16.4	14.2	11.1	9.4	7.2	6.6	6.4	7.0	8.8	12.5	14.8
	5% WB	MCDBR	20.2	22.0	20.9	20.6	17.3	16.3	15.7	14.4	12.8	13.5	16.1	18.4
		MCWBR	17.0	17.7	15.4	12.0	9.2	7.1	7.1	6.5	6.6	8.2	12.8	15.7
Clear-Sky Solar Irradiance	taub	0.308	0.321	0.347	0.391	0.434	0.483	0.511	0.505	0.407	0.356	0.324	0.313	
	taud	2.540	2.501	2.457	2.347	2.257	2.171	2.109	2.112	2.376	2.513	2.550	2.559	
	Ebn at Noon	282	291	292	283	271	256	248	247	268	275	273	272	
	Edh at Noon	25	29	33	39	43	47	50	49	36	28	24	23	
All-Sky Solar Radiation	RadAvg	743	984	1314	1674	1884	2023	1927	1738	1415	1124	840	662	
	RadStd	60	105	103	121	162	130	94	98	115	128	68	52	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (57 neighbors)	+0.81	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+270	+173
		+0.83	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+299	+191

Nomenclature: See separate page